
PH102: Physics-2

Electrodynamics

Lecture # 12

Electrostatics – Electric fields in matter – Boundary value problems with linear dielectrics and energy

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Previous lecture

- Any charge that is *not* a result of polarization is a free charge.
- In the presence of a dielectric, the Gauss's law is $\nabla \cdot \mathbf{D} = \rho_f$
- The curl of $\mathbf{D}$ is not always zero and so it cannot be expressed as the gradient of a scalar.
- The normal component of $\mathbf{D}$ is discontinuous by an amount σ_f at any boundary.

$$D_{above}^{\perp} - D_{below}^{\perp} = \sigma_f$$

- The parallel component of $\mathbf{D}$ is discontinuous at any boundary.

$$D_{above}^{\parallel} - D_{below}^{\parallel} = P_{above}^{\parallel} - P_{below}^{\parallel}$$

- For a linear dielectric medium,

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

$$\mathbf{D} = \epsilon \mathbf{E}$$

$$\epsilon = \epsilon_0 (1 + \chi_e)$$

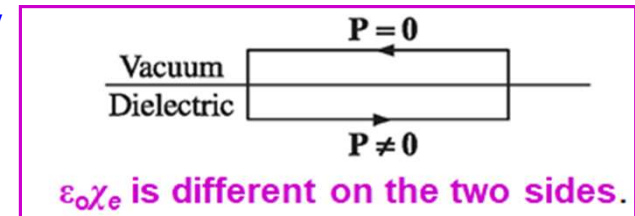
Curl of polarization across a boundary

If the entire medium is a **homogeneous linear dielectric**, then

$$\nabla \times \mathbf{D} = \nabla \times \mathbf{P} = 0, \quad \nabla \cdot \mathbf{D} = \rho_f$$

For a closed path that straddles the boundary between two different type of material then

$$\oint \mathbf{P} \cdot d\mathbf{l} \neq 0,$$



even though the integral of $\mathbf{E}$ around the same loop *must* be.

So, $\mathbf{D}$ can be found from the free charge just as though the dielectric were not there,

$$\nabla \times \mathbf{D} = \nabla \times (\epsilon_0 \mathbf{E} + \mathbf{P}) = \nabla \times (\epsilon_0 \mathbf{E}_{vac}) + (\nabla \times \mathbf{P}) = \nabla \times (\epsilon_0 \mathbf{E}_{vac}) + 0$$

$$\nabla \times \mathbf{D} = \nabla \times (\epsilon_0 \mathbf{E}_{vac})$$

$$\mathbf{D} = \epsilon_0 \mathbf{E}_{vac}$$

Therefore,

$$\mathbf{D} = \epsilon \mathbf{E} \Rightarrow \mathbf{E} = \frac{\mathbf{D}}{\epsilon} = \frac{\epsilon_0}{\epsilon} \mathbf{E}_{vac}$$

or,

$$\mathbf{E} = \frac{1}{\epsilon_r} \mathbf{E}_{vac}$$

When all space is filled with a homogeneous linear dielectric, the field everywhere is simply reduced by a factor of one over the dielectric constant.

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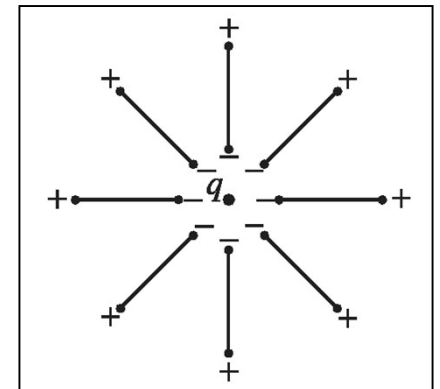
A free charge q is embedded in a large dielectric,

$$\mathbf{E} = \frac{\mathbf{E}_{vac}}{\epsilon_r} = \frac{1}{\epsilon_r} \left[\frac{1}{4\pi\epsilon_0} \left(\frac{q}{r^2} \right) \hat{\mathbf{r}} \right] = \frac{1}{4\pi\epsilon} \left(\frac{q}{r^2} \right) \hat{\mathbf{r}}$$

(that's ϵ , not ϵ_0)

The force on nearby charges is reduced accordingly.

The polarization of the medium **partially "shields" the charge**, by surrounding it with **bound charge of the opposite sign**.



Susceptibility tensor

A medium is said to be isotropic if its properties (such as susceptibility) are the same in all directions. Thus, following expression is a special case that holds for isotropic media.

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}$$

A *crystal* is generally easier to polarize in some directions than in others, and in this case above equation is replaced by the general linear relation

$$\left. \begin{aligned} P_x &= \epsilon_0(\chi_{e_{xx}} E_x + \chi_{e_{xy}} E_y + \chi_{e_{xz}} E_z) \\ P_y &= \epsilon_0(\chi_{e_{yx}} E_x + \chi_{e_{yy}} E_y + \chi_{e_{yz}} E_z) \\ P_z &= \epsilon_0(\chi_{e_{zx}} E_x + \chi_{e_{zy}} E_y + \chi_{e_{zz}} E_z) \end{aligned} \right\}$$

$\chi_{e_{ij}}$: components of the susceptibility tensor.

“Linear” just means that at any given point, and for $\mathbf{E}$ in a given direction, the components of $\mathbf{P}$ are proportional to $\mathbf{E}$ - the proportionality factor could vary with position and/or direction.

Boundary value problems with linear dielectrics

In a homogeneous linear dielectric, the bound charge density (ρ_b) is proportional to the free charge density (ρ_f),

$$\rho_b = -\nabla \cdot \mathbf{P} = -\nabla \cdot (\epsilon_0 \chi_e \mathbf{E}) = -\nabla \cdot \left(\epsilon_0 \frac{\chi_e}{\epsilon} \mathbf{D} \right) = -\left(\frac{\chi_e}{1 + \chi_e} \right) \rho_f$$

Unless free charge is actually embedded in the material, $\rho = 0$, and any net charge must reside at the surface.

Within such a dielectric, then, the potential obeys Laplace's equation.

Rewrite the boundary conditions in a way that makes reference only to the free charge.

$$D_{above}^\perp - D_{below}^\perp = \sigma_f \Rightarrow \epsilon_{above} E_{above}^\perp - \epsilon_{below} E_{below}^\perp = \sigma_f$$

or,

$$\epsilon_{above} \frac{\partial V_{above}}{\partial n} - \epsilon_{below} \frac{\partial V_{below}}{\partial n} = -\sigma_f$$

whereas the potential itself is continuous,

$$V_{above} = V_{below}$$

Capacitor filled with a linear dielectric medium

Electric field inside a dielectric medium is given by

$$\mathbf{E} = \frac{1}{\epsilon_r} \mathbf{E}_{vac}$$

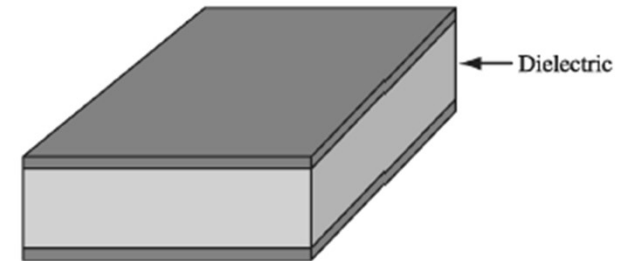
hence the potential difference V is

$$V = \frac{1}{\epsilon_r} V_{vac}.$$

The capacitance is therefore,

$$C = \frac{Q}{V} = \epsilon_r \frac{Q}{V_{vac}} = \epsilon_r C_{vac}$$

The capacitance is increased by a factor of the dielectric constant.



Energy in dielectric systems

The work to charge up a capacitor is $W = \frac{1}{2}CV^2$.

If the capacitor is filled with linear dielectric, its new capacitance is $C = \epsilon_r C_{\text{vac}}$

➤ the work necessary to charge a dielectric-filled capacitor is increased by the same factor.

The energy stored in any electrostatic system is

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau$$

And in the case of the dielectric-filled capacitor, it becomes

$$W = \frac{\epsilon_0}{2} \int \epsilon_r E^2 d\tau = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

Proof: Suppose, a free charge is added to a dielectric by an amount $\Delta\rho_f$ (the polarization will change and so the bound charge distribution but we're interested only in the work done), the work done on the incremental free charge

$$\Delta W = \int (\Delta\rho_f) V d\tau$$

$$\nabla \cdot \mathbf{D} = \rho_f \Rightarrow \Delta \rho_f = \nabla \cdot (\Delta \mathbf{D}), \quad \Delta \mathbf{D} \text{ is the resulting change in } \mathbf{D}$$

Therefore,
$$\Delta W = \int [\nabla \cdot (\Delta \mathbf{D})] V d\tau$$

Using
$$\nabla \cdot [(\Delta \mathbf{D})V] = [\nabla \cdot (\Delta \mathbf{D})]V + \Delta \mathbf{D} \cdot (\nabla V)$$

We get,
$$\Delta W = \int \nabla \cdot [(\Delta \mathbf{D})V] d\tau + \int (\Delta \mathbf{D}) \cdot \mathbf{E} d\tau$$

The divergence theorem turns the first term into a surface integral, which vanishes when we integrate over all surface. Therefore, the work done is

$$\Delta W = \int (\Delta \mathbf{D}) \cdot \mathbf{E} d\tau$$

For infinitesimal increments,

$$\frac{1}{2} \Delta(\mathbf{D} \cdot \mathbf{E}) = \frac{1}{2} \Delta(\epsilon E^2) = \epsilon(\Delta \mathbf{E}) \cdot \mathbf{E} = (\Delta \mathbf{D}) \cdot \mathbf{E}$$

The total work done, then, as we build the free charge up from zero to the final configuration, is

$$W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

What do we *mean* by "the energy of a system"?

Answer: It is the work required to assemble the system.

- Bring in all the charges (free *and* bound), one by one, with tweezers, and glue each one down in its proper final location, then the energy is

$$W = \frac{\epsilon_0}{2} \int E^2 d\tau$$

This will **not** include the work involved in stretching and twisting the dielectric molecules (the positive and negative charges as held together by tiny springs, it does not include the spring energy).

- Bring in the *free* charges, one by one, allowing the dielectric to respond as it sees fit, then the energy is

$$W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

The "spring" energy *is* included because as you move the free charge, you are automatically stretching those "springs."

Solved example 1

Suppose the entire region below the plane $z = 0$ is filled with uniform linear dielectric material of susceptibility χ_e . Calculate the force on a point charge q situated a distance d above the origin.

We must first calculate the bound charges.

The surface bound charge on the x - y plane is of opposite sign to q ,

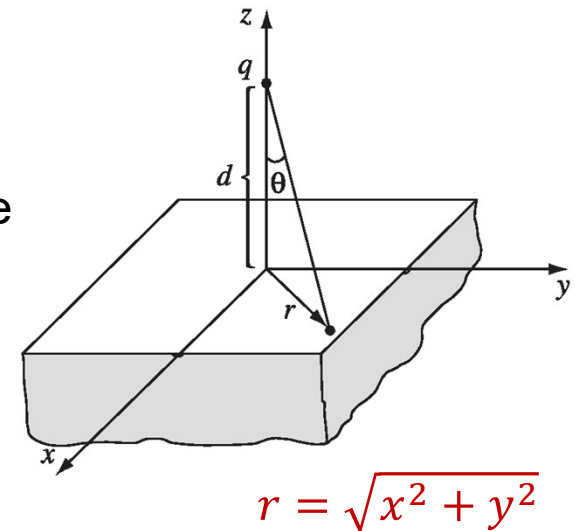
$$\sigma_b = \mathbf{P} \cdot \hat{n} = P_z = \epsilon_0 \chi_e E_z = \epsilon_0 \chi_e (E_q + E_b)$$

where E_z is the z -component of the total field just inside the dielectric, at $z = 0$. This field is due in part to q and in part to the bound charge itself.

There is no volume bound charge, because $\rho_f = 0$, $\rho_b = -\left(\frac{\chi_e}{1 + \chi_e}\right)\rho_f = 0$

The z component of the field due to q is

$$E_q = E \cos \theta = -\frac{1}{4\pi\epsilon_0} \frac{q}{(r^2 + d^2)} \cos \theta = -\frac{1}{4\pi\epsilon_0} \frac{qd}{(r^2 + d^2)^{3/2}}$$



The z component of the field of the bound charge is $E_b = -\frac{\sigma_b}{2\epsilon_0}$

Therefore, $\sigma_b = \epsilon_0 \chi_e E_z = \epsilon_0 \chi_e \left[-\frac{1}{4\pi\epsilon_0} \frac{qd}{(r^2 + d^2)^{3/2}} - \frac{\sigma_b}{2\epsilon_0} \right]$

and rearranging, we get

$$\sigma_b = -\frac{1}{2\pi} \left(\frac{\chi_e}{\chi_e + 2} \right) \frac{qd}{(r^2 + d^2)^{3/2}}$$

Compare this result with the image charge density in grounded conductor due to a point charge q (Lecture-9),

$$\sigma(x, y) = \frac{-qd}{2\pi[x^2 + y^2 + d^2]^{3/2}}$$

The total bound charge is therefore,

$$q_b = -\left(\frac{\chi_e}{\chi_e + 2} \right) q$$

Now, one can obtain the field of σ_b by direct integration, $\mathbf{E} = \frac{1}{4\pi\epsilon_0} \int \left(\frac{\hat{\mathbf{r}}}{r^2} \right) \sigma_b da$

Using the method of images, if we replace the dielectric by a single point charge q_b at the image position $(0, 0, -d)$, we get (in the region $z > 0$)

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z - d)^2}} + \frac{q_b}{\sqrt{x^2 + y^2 + (z + d)^2}} \right]$$

And, in the region $z < 0$, a charge $(q + q_b)$ at $(0, 0, d)$ yields the potential

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q + q_b}{\sqrt{x^2 + y^2 + (z - d)^2}} \right]$$

What about the boundary condition across the surface?

In the region $z > 0$,

$$\begin{aligned} \left. \frac{\partial V}{\partial z} \right|_{z=0^+} &= \frac{1}{4\pi\epsilon_0} \left[\frac{q(-1/2)2(z - d)}{[x^2 + y^2 + (z - d)^2]^{3/2}} + \frac{q_b(-1/2)2(z + d)}{[x^2 + y^2 + (z + d)^2]^{3/2}} \right] \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{(q - q_b)d}{[x^2 + y^2 + d^2]^{3/2}} \right] \end{aligned}$$

In the region $z < 0$,

$$\left. \frac{\partial V}{\partial z} \right|_{z=0^-} = \frac{1}{4\pi\epsilon_0} \left[\frac{(q + q_b)(-1/2)2(z - d)}{[x^2 + y^2 + (z - d)^2]^{3/2}} \right] = \frac{1}{4\pi\epsilon_0} \left[\frac{(q + q_b)d}{[x^2 + y^2 + d^2]^{3/2}} \right]$$

Therefore,

$$\begin{aligned} \left. \frac{\partial V}{\partial z} \right|_{z=0^+} - \left. \frac{\partial V}{\partial z} \right|_{z=0^-} &= \frac{1}{4\pi\epsilon_0} \left[\frac{-2q_b d}{[x^2 + y^2 + d^2]^{3/2}} \right] \\ &= \frac{1}{4\pi\epsilon_0} \left[\frac{-2d}{[x^2 + y^2 + d^2]^{3/2}} \right] \left(\frac{-\chi_e}{\chi_e + 2} \right) q \\ &= \frac{1}{2\pi\epsilon_0} \left[\frac{qd}{[x^2 + y^2 + d^2]^{3/2}} \right] \left(\frac{\chi_e}{\chi_e + 2} \right) = \frac{-\sigma_b}{\epsilon_0} \end{aligned}$$

$$-\epsilon_0 \left(\left. \frac{\partial V}{\partial z} \right|_{z=0^+} - \left. \frac{\partial V}{\partial z} \right|_{z=0^-} \right) = -\frac{1}{2\pi} \left(\frac{\chi_e}{\chi_e + 2} \right) \frac{qd}{(x^2 + y^2 + d^2)^{3/2}} = \sigma_b$$

The force on q is $\mathbf{F} = \frac{1}{4\pi\epsilon_0} \frac{qq_b}{(2d)^2} \hat{\mathbf{z}} = -\frac{1}{4\pi\epsilon_0} \left(\frac{\chi_e}{\chi_e + 2} \right) \frac{q^2}{4d^2} \hat{\mathbf{z}}$

There are at least two "rules of the game":

- You must never put an image charge into the region where you're computing the potential. For the potential in the region $z > 0$, image charge q_b is at $z = -d$; In the region $z < 0$, the image charge ($q + q_b$) is at $z = +d$.
- The image charges must add up to the correct total in each region. (That's how we knew to use q_b , to account for the charge in the region $z < 0$, and ($q + q_b$) to cover the region $z > 0$.)

Solved example 2

Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z \geq 0$, $\epsilon_{r1} = 4$ and for $z \leq 0$, $\epsilon_{r2} = 3$. A uniform electric field $\mathbf{E}_1 = 5\hat{x} - 2\hat{y} + 3\hat{z}$ exists for $z \leq 0$. Find

(a) $\mathbf{E}_2$ for $z \geq 0$.

(b) The angles $\mathbf{E}_1$ and $\mathbf{E}_2$ make with the interface.

We have to calculate $\mathbf{E}$, so express boundary conditions in terms of $\mathbf{E}$.

$$D_2^\perp - D_1^\perp = \sigma_f = 0 \quad (\text{No free charge})$$

Therefore,

$$\epsilon_{r1} E_1^\perp = \epsilon_{r2} E_2^\perp$$

and

$$\mathbf{E}_1^\parallel = \mathbf{E}_2^\parallel$$

Let

$$\mathbf{E}_2 = a\hat{x} + b\hat{y} + c\hat{z}$$

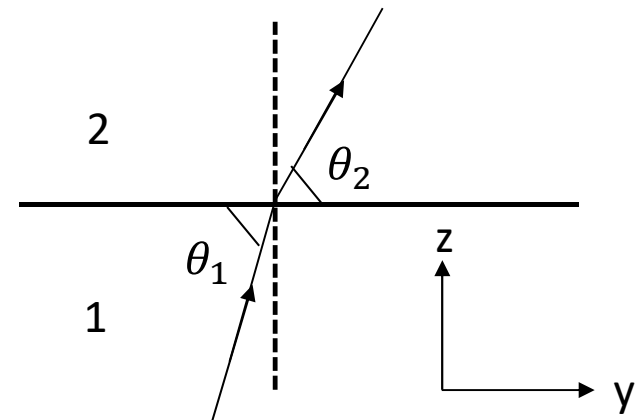
Since x-y plane is the parallel plane, hence

$$\mathbf{E}_1^\parallel = 5\hat{x} - 2\hat{y} = \mathbf{E}_2^\parallel$$

For the perpendicular direction, $4(3) = 3c \Rightarrow c = 4$

Therefore,

$$\mathbf{E}_2 = 5\hat{x} - 2\hat{y} + 4\hat{z}$$

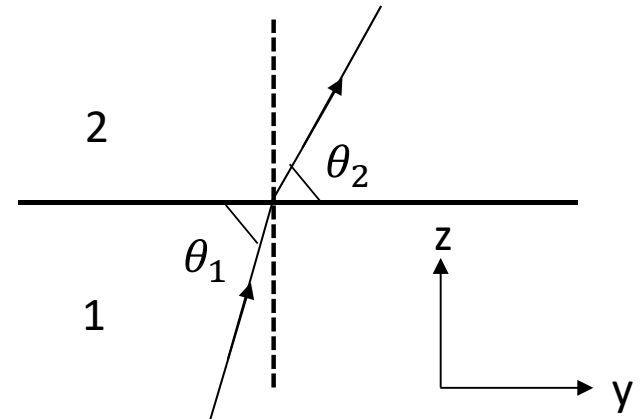


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From the figure,

$$\tan \theta_1 = \frac{E_1^\perp}{E_1^\parallel} = \frac{3}{\sqrt{29}} \Rightarrow \theta_1 = 29.1^\circ$$

$$\tan \theta_2 = \frac{E_2^\perp}{E_2^\parallel} = \frac{4}{\sqrt{29}} \Rightarrow \theta_2 = 36.6^\circ$$



Solved example 3

A sphere of radius R is filled with material of dielectric constant ϵ_r and uniform embedded free charge ρ_f . What is the energy of this configuration?

Given free charge density,

$$\mathbf{D}(r) = \begin{cases} \frac{\rho_f}{3} \mathbf{r} & (r < R), \\ \frac{\rho_f}{3} \frac{R^3}{r^2} \hat{\mathbf{r}} & (r > R). \end{cases} \quad \text{therefore} \quad \mathbf{E}(r) = \begin{cases} \frac{\rho_f}{3\epsilon_0\epsilon_r} \mathbf{r} & (r < R), \\ \frac{\rho_f}{3\epsilon_0} \frac{R^3}{r^2} \hat{\mathbf{r}} & (r > R). \end{cases}$$

$$W_1 = \frac{\epsilon_0}{2} \left[\left(\frac{\rho_f}{3\epsilon_0\epsilon_r} \right)^2 \int_0^R r^2 4\pi r^2 dr + \left(\frac{\rho_f}{3\epsilon_0} \right)^2 R^6 \int_R^\infty \frac{1}{r^4} 4\pi r^2 dr \right] \quad W = \frac{\epsilon_0}{2} \int E^2 d\tau$$

$$= \frac{2\pi}{9\epsilon_0} \rho_f^2 R^5 \left(\frac{1}{5\epsilon_r^2} + 1 \right). \quad \text{Electrostatic energy}$$

$$W_2 = \frac{1}{2} \left[\left(\frac{\rho_f}{3} \right) \left(\frac{\rho_f}{3\epsilon_0\epsilon_r} \right) \int_0^R r^2 4\pi r^2 dr + \left(\frac{\rho_f R^3}{3} \right) \left(\frac{\rho_f R^3}{3\epsilon_0} \right) \int_R^\infty \frac{1}{r^4} 4\pi r^2 dr \right] \quad W = \frac{1}{2} \int \mathbf{D} \cdot \mathbf{E} d\tau$$

$$= \frac{2\pi}{9\epsilon_0} \rho_f^2 R^5 \left(\frac{1}{5\epsilon_r} + 1 \right). \quad \text{Total energy}$$

$W_1 < W_2$ because W_1 does not include the energy involved in stretching the molecules.

Questions?